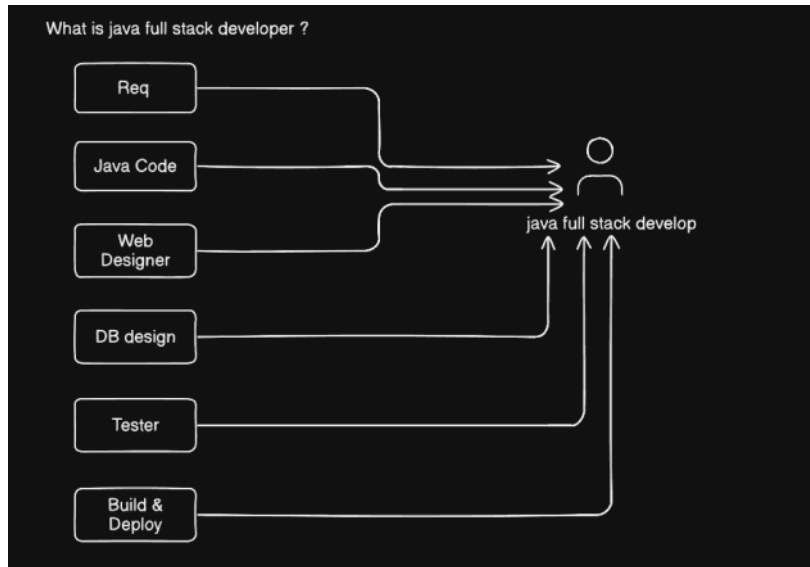


Introduction Java

07 June 2026 09:05

What is java developer ?



Java Application Architectures:

1. Standalone Applications : An application that is installed on single isolated machine and executed it's action entirely within same computer /machine
 - a. Core Technology stack : Core java (J2SE - java 2 standard edition)

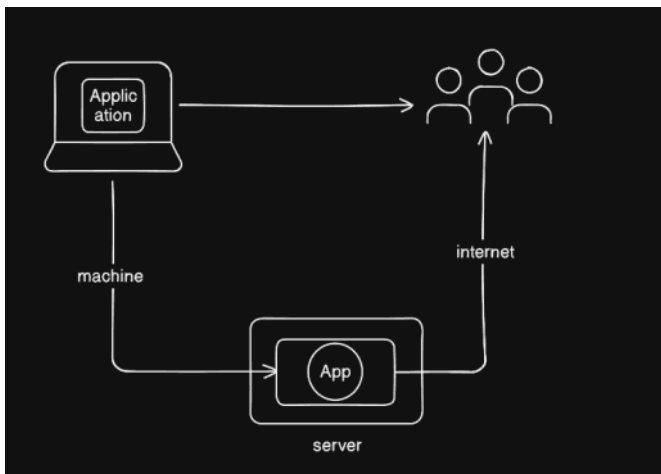
Developer perspective :

1. No HTML:

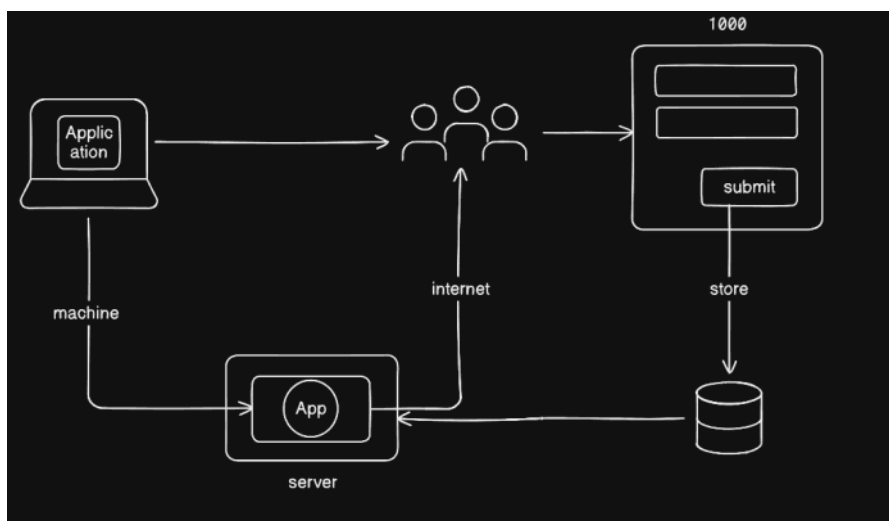
```
Code sample
-----
INVOICE
-----
Date      1/16/2024 7:57 AM
Customer  John
Drink     Espresso
Notes     Added Espresso Shots x2, Caramel Syrup
          Pumps x3, Mocha Drizzle, Extra
Total     $6.50
```

2. No Server environment: isko run karne ke liy kisi external machine ki jarurat nhi ha

What is server ?



3. NO Database storage:



2. Web Application

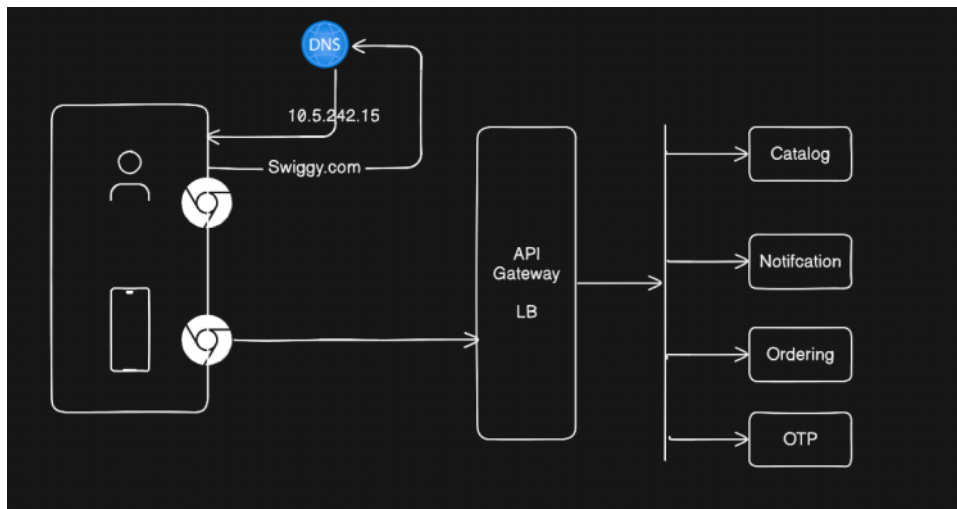
An application that executes with Web environment or internet environment is known as web application

Tech Stack:

1. JDBC
2. Servlet
3. JSP

3. Enterprise Application

An application that execute within a distributes environment and relies heavily on advance infra services like security, load balancer, clustering Is knows EA or EDP



Core Concepts of Software Development:

1. Programming language: A language that provides syntax rule, structure and concepts
It tells system how to allocate memory, manage data types, evaluate condition, execute conditions

Eg: lang=vocabulary of English

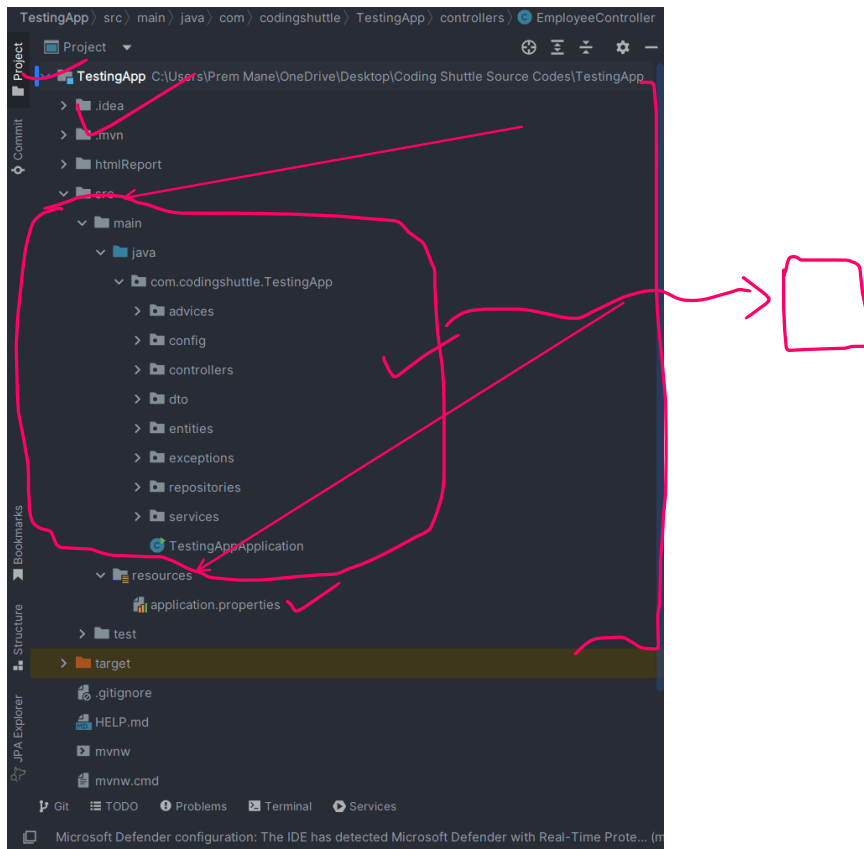
Core java : provide basic syntax, classes, interfaces

2. Technology: The process of converting theoretical knowledge and language concept into Realtime functionality or application development is known as technology

Technology act as bridge/specification that utilize a language to solve specific problem(like web communication, database connectivity etc)

3. Framework: is a semi- built, ready-constructed architectural structure available for application development.

It provides pre-written (boilerplate code), standard design patterns and lib allowing developers to focus on business login instead of setting up everything scratch.



Fundamentals of programming & Code Execution lifecycle

Har ek developer ko pata hona chahiye ki backend par code likhne se lekar uske run hoke tak system ke ander kya kya steps hote ha

Program: [set of instruction given to machine to perform a specific task](#)

Programming: The overall process used to construct program. It specifically means converting Analysis and design into readable source code.

Programmer: The person who writes the program/source code.

Code Lifecycle stages:

Jab ek programmer source code likhta ha to use respective language extension k sath save krta ha

eg: Test.c, Test.cpp, Test.py

Eg: Test.java

Source code -----> compilation process -----> execution process -----> output

Stage A: compilation process

Process of [checking whether the constructed program follow the strict syntax rules and grammar](#) of that specific programming language is known as compilation process.

The Compiler: lang like C, C++ and java use dedicated s/w component called compiler to scan and verify the code you have written.

Stage B : Execution Process:

[The process of running the compiled code to verify the required and expected output](#) generated or not is known as execution process.

execution steps of C lang:
